JFORCE SOLUTIONS

SQL DEVELOPER – STORED PROCEDURE INTERVIEW TASK

Level: Junior / R1 | Duration: 2–3 Hours

1. Objective

Create and demonstrate SQL Stored Procedures for an Employee Management and Payroll database. The task is designed to evaluate the candidate's understanding of procedure creation, parameters, conditional logic, calculations, aggregation, and basic error handling.

2. Database Setup

Create an employees table with the following columns:

Column	Data Type	Requirement
employee_id	INT	Primary Key
employee_name	VARCHAR	Employee name
department	VARCHAR	Department name
designation	VARCHAR	Job designation
basic_salary	DECIMAL	Basic salary
joining_date	DATE	Date of joining
status	VARCHAR	Active / Inactive

Insert at least 10 sample employees across different departments and salary ranges.

3. Task 1 – Employee Salary Report Procedure

Create a stored procedure named GetEmployeeSalaryReport. The procedure must accept department_name and minimum_salary as input parameters.

The procedure should return employees who:

- Belong to the specified department.
- Have a basic salary greater than or equal to the specified minimum salary.
- Have status = 'Active'.

Output: Employee ID, Employee Name, Department, Designation, Basic Salary, Joining Date.

Example execution:

```
CALL GetEmployeeSalaryReport('IT', 50000);
```

4. Task 2 – Employee Salary Calculation Procedure

Create a stored procedure named CalculateEmployeeSalary. It should accept employee_id as an input parameter and calculate the employee's salary components using the following rules:

- HRA = 20% of Basic Salary
- DA = 10% of Basic Salary
- PF = 12% of Basic Salary
- Net Salary = Basic Salary + HRA + DA - PF

Output: Employee Name, Basic Salary, HRA, DA, PF, Net Salary.

Example execution:

```
CALL CalculateEmployeeSalary(101);
```

5. Task 3 – Salary Grade Procedure

Create a stored procedure named EmployeeSalaryGrade. It should accept employee_id and assign a salary grade using the following rules:

Basic Salary	Grade
Below ₹30,000	C
₹30,000 – ₹60,000	B
Above ₹60,000	A

Output: Employee Name, Basic Salary, Salary Grade.

6. Bonus Task – Department Summary

Create a procedure named GetDepartmentSummary. It should accept a department name and return:

- Total number of employees
- Average salary
- Highest salary
- Lowest salary
- Total salary expenditure

Bonus: Handle the situation where the requested department does not exist and display an appropriate message.

7. Technical Expectations

- Use appropriate input parameters.
- Use meaningful procedure and variable names.
- Use IF/CASE logic wherever appropriate.
- Use aggregate functions where required.
- Include basic error or validation handling where applicable.
- Format the SQL script clearly and add comments for important sections.

8. Submission Requirements

1. database.sql – Table creation and sample data.
2. procedures.sql – All stored procedures.
3. README.md – Brief explanation of the solution.
4. GitHub repository link.
5. 2–5 minute demo video showing procedure creation, execution, and output.

9. Evaluation Criteria Submit the GitHub link and demo video to: careers@jforcesolutions.com

Area		Marks
Database and table design		10
Stored procedure syntax		20
Parameters		15
IF/CASE logic		15
Salary calculations		15
Error/validation handling		10
Code quality and documentation		5
Explanation and demo		10
Total		100

Good luck!